Semaev first-fall replay and Boolean-parity correction

2026-10-22 — scoped validation of `crypto-autoresearcher` main `4db6ed75`

The original eight-cell summary reproduces byte for byte: 70 of 80 Semaev draws first fall at degree two, ten at degree three. Its 200 support-matched null-control grades also reproduce. The archived `fall_profile` multiplies Boolean polynomials by collecting OR-combined monomial masks in a *set*. Equal monomials should cancel over $\mathbb{F}_2$. With that parity corrected on the same frozen equations and seeds, the counts are **68 at degree two and 12 at degree three**; two nonzero-target draws change grade. Twelve of 200 null grades change. This is a source-code finding, not a new cryptanalytic timing or attack-cost measurement.

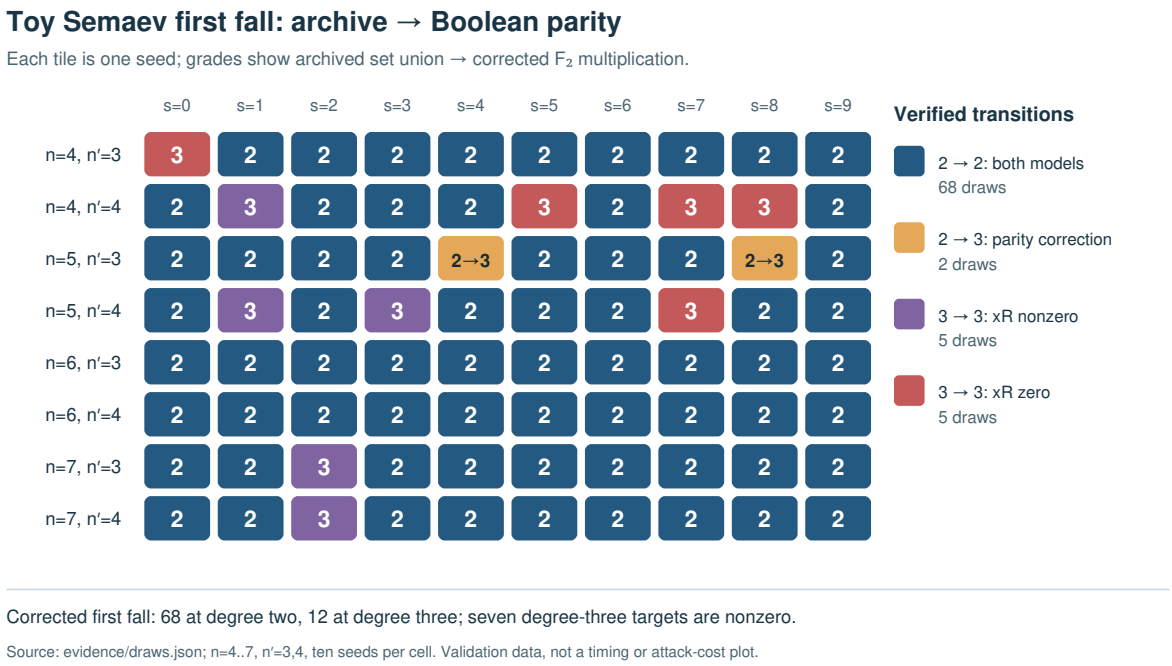


Figure: each tile shows archived → parity-corrected first-fall grade for one exact Semaev seed. Source: `evidence/draws.json`; editable renderer: `render.py`. The historical 70/10 graph remains an accurate picture of the archived table, while this chart shows the corrected Boolean result.

A direct multiplication witness

For $n = 4$, $n' = 4$, seed index $s = 1$, multiply the first component equation by u_0 (mask 1). The archived set product retains masks 1, 17, and 129; all three cancel in the $\mathbb{F}_2$ product. On the complete 256-assignment truth table, the archived product disagrees with evaluation of $f\,u_0$ in 64 assignments, and the parity-corrected product agrees in all 256. The replay checker asserts this witness, compares 21,760 finite-field products and 12,800 direct S_3 assignments, and independently computes both rank profiles on all 80 Semaev and 200 null systems. All archived distributions match the frozen table.

Finite first-fall counterexample

The same $n = 4, n' = 4, s = 1$ system has $x_R = 1$, basis $[9, 3, 2, 12]$, and field modulus $X^4 + X + 1$. Its trace residual $u_0 + u_3 + w_0 + w_3$ is already the fourth input equation. Under proper Boolean parity, $\dim V_1 = 1$ and $\dim(V_2 \cap R_{\le 1}) = 1$, hence new_falls(2) $= 0$. Also $\dim V_2 = 11$, $\dim V_3 = 53$ and $\dim(V_3 \cap R_{\le 2}) = 12$, giving new_falls(3) $= 1$. Thus first fall is three even with $x_R \neq 0$. The archived set rule instead reports $\dim V_3 = 59$ and seven new falls at three; its grade happens to agree for this draw.

Proof boundary and exact exceptions

The existing `DegreeTwoFall.lean` proves quadratic trace cancellation under $x_R \neq 0$, not a nonzero quotient class. The corrected finite example contradicts a uniform degree-two conclusion for this toy Boolean model; it does not refute the narrower Lean identity or imply a solving-degree or ECDLP cost result. A new finite Lean certificate now checks the transcribed four-equation system: its kernel-reduced ranks give no new fall at degrees one or two and one new fall at degree three. The audited theorems use only `propext`. The generic proof that its executable `rankOn` equals vector-space dimension, and a Lean derivation of the equations from abstract S_3, remain open. The frozen target and historical measurement were not edited.

n	n'	seed index s	x_R	archived → corrected
4	4	1	1	$3 \to 3$
5	3	4	8	$2 \to 3$
5	3	8	8	$2 \to 3$
5	4	1	19	$3 \to 3$
5	4	3	19	$3 \to 3$
7	3	2	125	$3 \to 3$
7	4	2	113	$3 \to 3$

These seven corrected degree-three draws all have $x_R \neq 0$. Five other degree-three draws have $x_R = 0$ and lie outside the Lean theorem premise. Among the 75 nonzero-target draws, the trace residual is a new class modulo V_1 in 67 corrected degree-two cases; one other degree-two case has an old trace class but another new relation. Six degree-three cases have a nonzero trace residual already in V_1; one has a zero trace relation.

Reproduction and remaining work

The original script was run from a scratch directory and its JSON compared byte for byte with the archive (SHA-256 `64f6e34ceb16bd81…54 333c99e7`). The Python 3.12 checker, `replay_ffd_semaev_20261022.py`, records every exact equation system, source hash, invocation, archived and corrected rank profile in `evidence/draws.json`; the full methodology and limits are in `report.md`. The pinned Lean source is `BooleanParityFiniteCertificate.lean`; its source hash, command and axiom output are in `evidence/lean-certificate.json` and `evidence/lean-certificate.log`. The four Lean equation lists were checked byte-for-byte against the per-draw record; see `evidence/lean-literal-check.log`. The pinned full workspace also built 2,012 jobs, including this module, and audited 475 theorem declarations with standard axioms only; see `evidence/full-workspace-replay.json`. The repository's single-target verifier returned `machine_verified` locally; its exact receipt and scratch-root hash binding are in `evidence/`. Hosted CI and independent semantic review remain pending. The next formal obligation is to prove the rank semantics and equation derivation, then correct the broader frozen claim additively through review. No official research-state transition was made.